Q. One: For the process $X(t) = \cos\left(t + \frac{\theta}{2}\right)$ where $\theta \sim U(0, 2\pi)$, we have $E(X(t)) =$

A) $-\frac{4\,si\ (t)}{\pi}$ B) $-\frac{2\sin(t)}{\pi}$ C) 0 D) 3π E) None

Q. Two: Consider a mean-ergodic process $X(t)$ with no periodic component and $R_{XX}(\tau) = a^2 + be^{-|\tau|}$ where a, b are positive constants. If $E(X(t)) > 0$, then

A) The mean and variance of the process are represented by a and b ; respectively
B) The mean and variance of the process are represented by b and a ; respectively
C) The mean and variance of the process are represented by b and $-a$; respectively
D) The mean and variance of the process are represented by $-a$ and b ; respectively
E) None

Q. Three: For a random process $X(t) = 2\,A\cos(t)$ where A is a random variable with mean 0 and variance 1 . We have $E(X^2(t)) =$

A) $\sin(t)$ B) $2\sin(t)$ C) $\sin^2(t)$ D) $2\sin^2(t)$ E) None

Q. Four: For $X \sim Poisson(m)$ with $P(X = 0) = e^{-2}$; then $P(X > 1) =$

A) e^{-2} B) $1 - 2e^{-2}$ C) $1 - 3e^{-2}$ D) $1 - e^{-2}$ E) None

Q. Five: Orthogonal processes $X(t)$, $Y(t)$ with $R_{XY}(\tau) = 2 + a$, then $a =$

A) -2 B) -3 C) 2 D) 3 E) None

Q. Six: Jointly wide-sense stationary processes $X(t)$, $Y(t)$ with $R_{XY}(\tau) = 2 + \bar{X}\,\bar{Y}$, then $C_{XY}(t, t+\tau) =$

A) -3 B) 2 C) -2 D) 3 E) None

Q. Seven: Let X be a random variable and $Y = \bar{X} - X$. The value of $E(XY)$ equals

A) 0 B) $-\mathrm{Var(X)}$ C) 1 D) Var(X) E) None

Q. Eight: For $X \sim N(0, \sqrt{2})$ then $E(X) + std(X) =$

A) 1 B) $\sqrt[3]{2}$ C) $\sqrt{2}$ D) $\sqrt[4]{2}$ E) None

Q. Nine: The variance of the process $X(t) = 3A$ where $A \sim U(0,1)$ is equal to

A) $\frac{1}{3}$ B) $\frac{1}{6}$ C) $\frac{3}{4}$ D) $\frac{1}{4}$ E) None

1) $f(\theta) = \begin{cases} \frac{1}{2\pi} & , 0 \le x \le 2\pi \\ 0 & , o.w \end{cases}$

$$E(x(t)) = \int_{-\infty}^{\infty} x(t) \cdot f(\theta)\, d\theta = \frac{1}{2\pi} \int_{0}^{2\pi} \cos\left(t + \frac{\theta}{2}\right) d\theta$$

$$= \frac{2}{2\pi} \sin\left(t + \frac{\theta}{2}\right) \Big|_{0}^{2\pi} = \frac{1}{\pi} \sin(t+\pi) - \sin t$$

$$= -\frac{2\sin t}{\pi}$$

[B]

Remember: $\sin(a+b) = \sin a \cos b + \cos a \sin b$

$\sin(t+\pi) = \sin t \cos\pi + \cos t \sin\pi$ (cos π = −1, sin π = 0)

$= -\sin t$

2) $\bar{X} = \sqrt{\lim_{|\tau| \to \infty} R_{XX}(\tau)} = \sqrt{a^2} = a$

$$Var(X) = R_{XX}(\tau = 0) - \bar{X}^2$$

$$= a^2 + b - a^2 - b$$

[A]

3) $Var(A) = E(A^2) - (EA)^2 \longrightarrow E(A^2) = 1$

$E(x^2(t)) = E(4A^2\cos^2 t) = 4\cos^2 t$

[E]

4) $P(X>1) = 1-\left(P(0)+P(1)\right) = 1-\left(e^{-2}+\frac{2e^{-2}}{1!}\right)$

$m=2$

$= 1-3e^{-2}$ (C)

5) $R_{XY}(\tau) = 2+a=0 \longrightarrow a=-2$ (A)

6) $C_{XY}(t,t+\tau) = R_{XY}(t,t+\tau) - \bar{X}\bar{Y}$

$= 2+\bar{X}\bar{Y}-\bar{X}\bar{Y} = 2$ (B)

7) $XY = XEX - X^2$

$E(XY) = (EX)^2 - E(X^2) = -Var(X)$ (B)

8) $Var(X) = \sqrt{2} \longrightarrow std(X) = \sqrt[4]{2}$ (D)

$EX = 0$

9) $f(A) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & o.w \end{cases}$

$E(x^2(t)) = \int_0^1 9A^2 dA = \frac{9}{3}A^3\Big]_0^1 = 3$

$\left(E(x(t))\right)^2 = \left(\int_0^1 3A\,dA\right)^2 = \left(\frac{3}{2}A^2\Big]_0^1\right)^2 = \frac{9}{4}$

$Var(X) = 3-\frac{9}{4} = \frac{12-9}{4} = 3/4$ (C)

Q. Ten: Consider the joint pdf $f(x,y) = x$ when $0 < x < 2$, $0 < y < 1/2$ and zero otherwise. Then

A) $f_X(x) = \frac{x}{2}$ and $f_Y(y) = 1$ B) $f_X(x) = x$ and $f_Y(y) = 2$

C) $f_X(x) = \frac{x}{2}$ and $f_Y(y) = 2$ D) $f_X(x) = x$ and $f_Y(y) = 1$ E) None

Q. Eleven: Consider the joint pdf $f(x,y) = 1$ when $0 < x < 2$, $0 < y < 1/2$ and zero otherwise. Then $E(XY) =$

A) $\frac{1}{3}$ B) $\frac{1}{4}$ C) $\frac{3}{4}$ D) $\frac{1}{6}$ E) None

Q. Twelve: For the process $X(t) = \cos(wt)$, $w \sim U(0,b)$ if $E(bX(t)) = \frac{\sin(t)}{t}$; then the value of the positive constant b equals

A) 1 B) -1 C) 2 D) -2 E) None

Q. Thirteen: Which of the following is wrong

A) The binomial distribution is a discrete probability distribution
B) The Poisson's distribution is a discrete probability distribution
C) The Normal distribution is a continuous probability distribution
D) The uniform distribution is a discrete probability distribution
E) None

Q. Fourteen : For some random variables X, Y, if $Cov(X,Y) = 2$ and $\sigma_X = \sigma_Y = 2$, then $Corr(X,Y)$ equals

A) 0.5 B) -0.5 C) 1 D) -1 E) None

Q. Fifteen: A fair coin is tossed 5 times. The probability of getting no heads is

A) $\frac{1}{2}$ B) $\frac{1}{64}$ C) $\frac{3}{16}$ D) $\frac{1}{32}$ E) None

Q. Sixteen: A website gets an average of 2 visits per hour. The probability that exactly three people visit the site in a given hour is

A) $\frac{4e^{-2}}{3}$ B) $\frac{2e^{-2}}{3}$ C) e^{-2} D) $2e^{-2}$ E) None

10) $f_X(x) = \int_0^{1/2} x \, dy = \frac{x}{2}$

$f_Y(y) = \int_0^2 x \, dx = \frac{x^2}{2}\Big]_0^2 = \frac{y}{2} = 2$ [C]

11) $E(XY) = \int_0^{1/2} x \left(\int_0^2 y \, dy \right) dx = \int_0^{1/2} 2x \, dx = x^2\Big]_0^{1/2} = \frac{1}{4}$

$2 = \frac{y^2}{2}\Big]_0^2$ ← [B]

12) $f(w) = \begin{cases} \frac{1}{b}, & 0 \le w \le b \\ 0, & o.w \end{cases}$

$b\,E(x(t)) = \frac{b}{b}\int_0^b \cos(wt)\,dw = \frac{\sin(wt)}{t}\Big]_0^b$

$\frac{\sin(bt)}{t} = \frac{\sin t}{t}, \quad b = 1$ [A]

13) Discrete → Binomial
→ Poisson

[D]

Continuous → Normal
→ Uniform

14) $Corr(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} = \frac{2}{2 \cdot 2} = \frac{1}{2}$ [A]

15) $n = 5$, $p = q = 0.5$

$X \sim B(5, 0.5)$ [D]

$P(X=5) = \binom{5}{5} (0.5)^5 (0.5)^{5-5} = \frac{1}{2^5} = \frac{1}{32}$

16) $X \sim Poisson(2)$

$P(X=3) = \frac{e^{-2} 2^3}{3!} = \frac{8e^{-2}}{3 \cdot 2} = \frac{4e^{-2}}{3}$ [A]

$$f(A) = \begin{cases} 1, & 0 \le A \le 1 \\ 0, & o.w \end{cases}$$

Q. Seventeen: For processes $X(t) = A\cos(t)$ and $Y(t) = A$ where $A \sim U(0,1)$; calculate the Cross-covariance

$$C_{xy}(t, t+\tau) = R_{xy}(t, t+\tau) - E(x(t)) \cdot E(Y(t+\tau))$$

$$= \int_0^1 A^2 \cos t \, dA - \left(\int_0^1 A \cos t \, d \right)\left(\int_0^1 A \, dA \right)$$

$$= \frac{A^3}{3}\cos t \Big]_0^1 - \left(\frac{A^2}{2}\cos t \Big]_0^1 \right)\left(\frac{A^2}{2} \Big]_0^1 \right)$$

$$= \frac{\cos t}{3} - \frac{\cos t}{4} = \cos t\left(\frac{1}{3} - \frac{1}{4}\right) = \frac{\cos t}{12}$$

$$\frac{4-3}{12}$$

Q. Eighteen : Determine whether the process $X(t) = 2A\cos(t)$ where $A \sim U(0,1)$ is first order stationary or not

$$f(A) = \begin{cases} 1, & 0 \le A \le 1 \\ 0, & o.w \end{cases}$$, to be 1st order : $E(x(t)) = \bar{X} \equiv$ constant

$$E(x(t)) = \int_{-\infty}^{\infty} x(t) \cdot f(A) \, dA = \int_0^1 2A \cos t \, dA$$

$$= \frac{2A^2}{2}\cos t \Big]_0^1 = \cos t \neq \bar{X}$$

therefore: it's NOT 1st order stationary.